

CS 3005

Theory of Automata

Lecture 22

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FAST NUCES CFD

2 PDA or two Stack PDA

- Let us define a two-stack pushdown automaton as follows

$$M = (K, \Sigma, \Gamma, \Delta, S, F)$$

- K is a finite set of states,
- Σ is an alphabet (the input symbols),
- Γ is an alphabet (the stack symbols),
- $S \in K$ is the start state,
- $F \subseteq K$ is the set of final states, and Finite set of transition from one state to another state

2PDA

- As we have two stacks for storage purpose, so change the name of POP and PUSH stack accordingly.
- POP_1 will pop from the first stack
- POP_2 will pop from the second stack
- PUSH_1 will push into first stack
- PUSH_2 will push into second stack

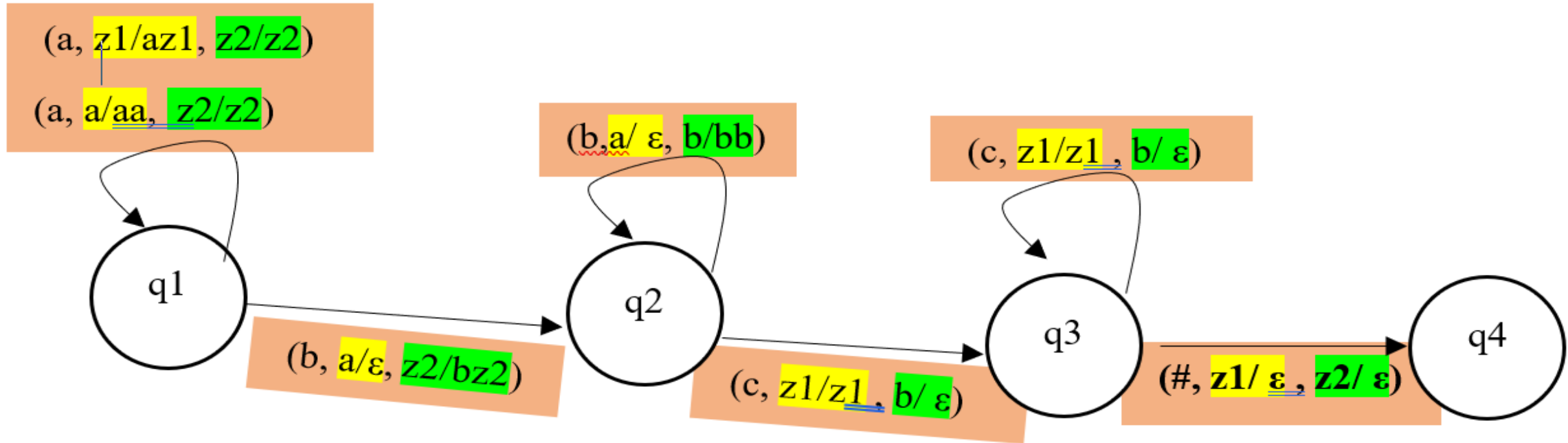
Guess the Language $a^n b^n c^n$

Input Tape						
a	a	b	b	c	c	#

z1

z2

State Machine

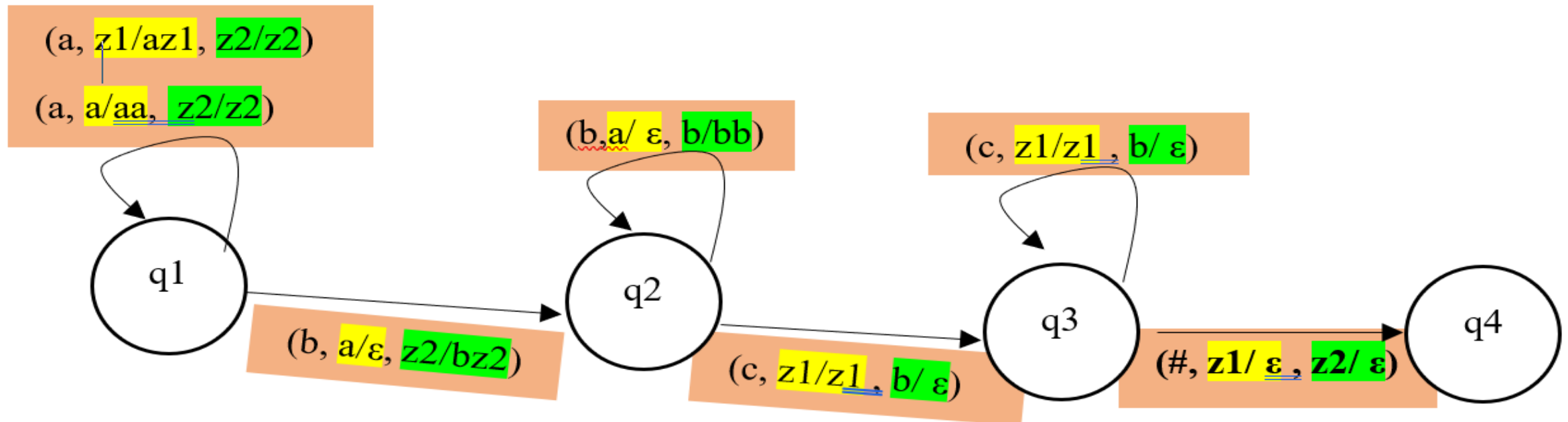


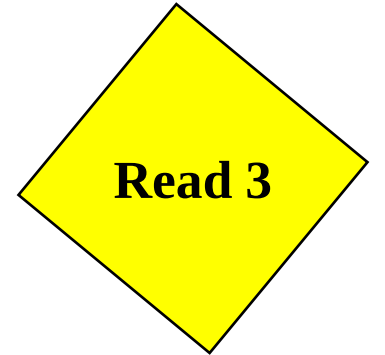
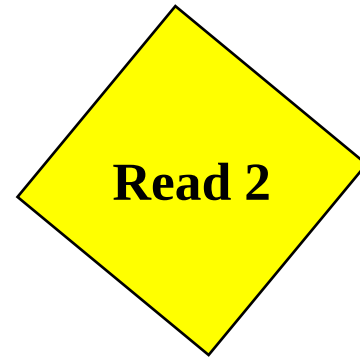
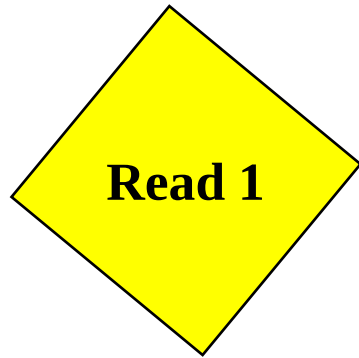
Transition Function

States	Input string	Stack 1	Stack 2
q1	aabbcc#	z1	z2
q1	abbcc#	az1	z2
q2	bbcc#	aaz1	z2
q2	bcc#	az1	bz2
q2	cc#	z1	bbz2
q3	c#	z1	bz2
q3	#	z1	z2
q4	-----	-----	-----

Convert state machine in Flow Machine

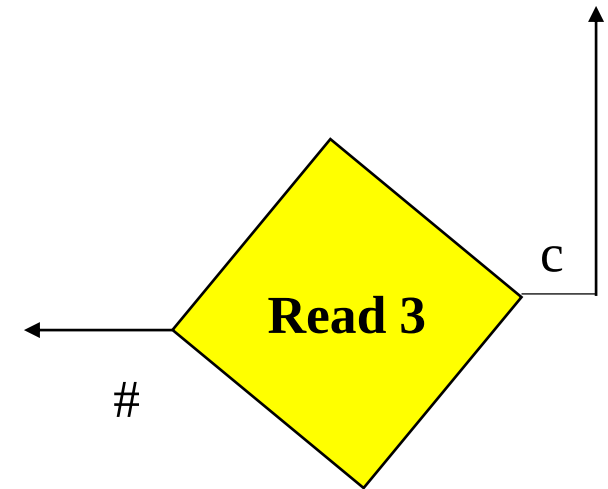
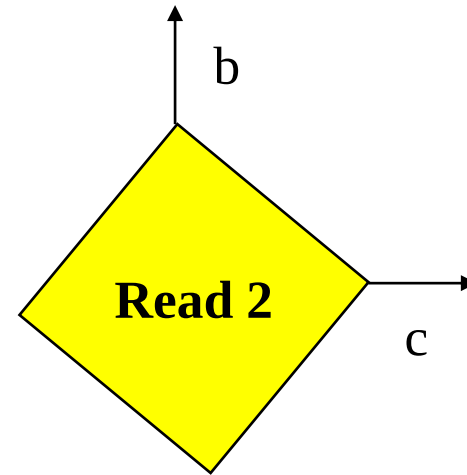
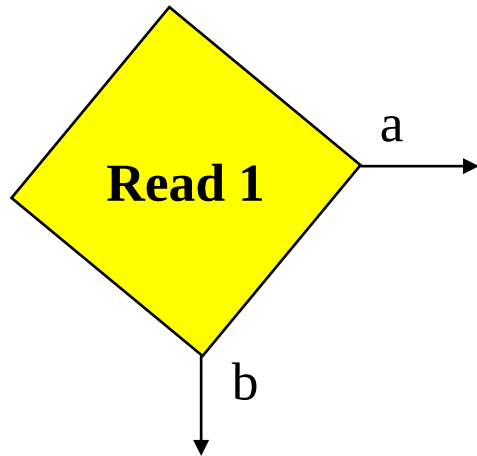
- No of states except final states = 3
- Final state should be as accept state



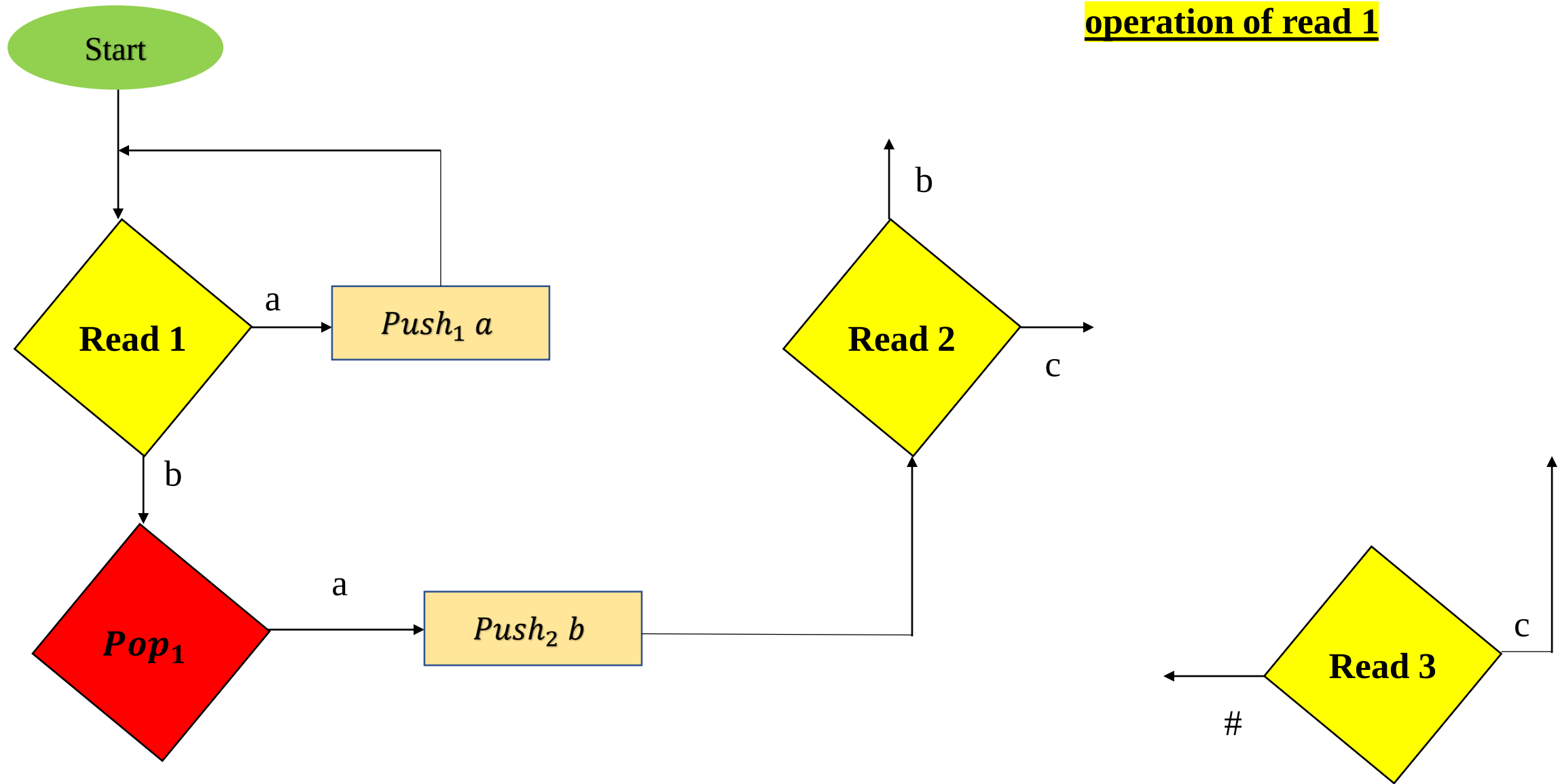


Start

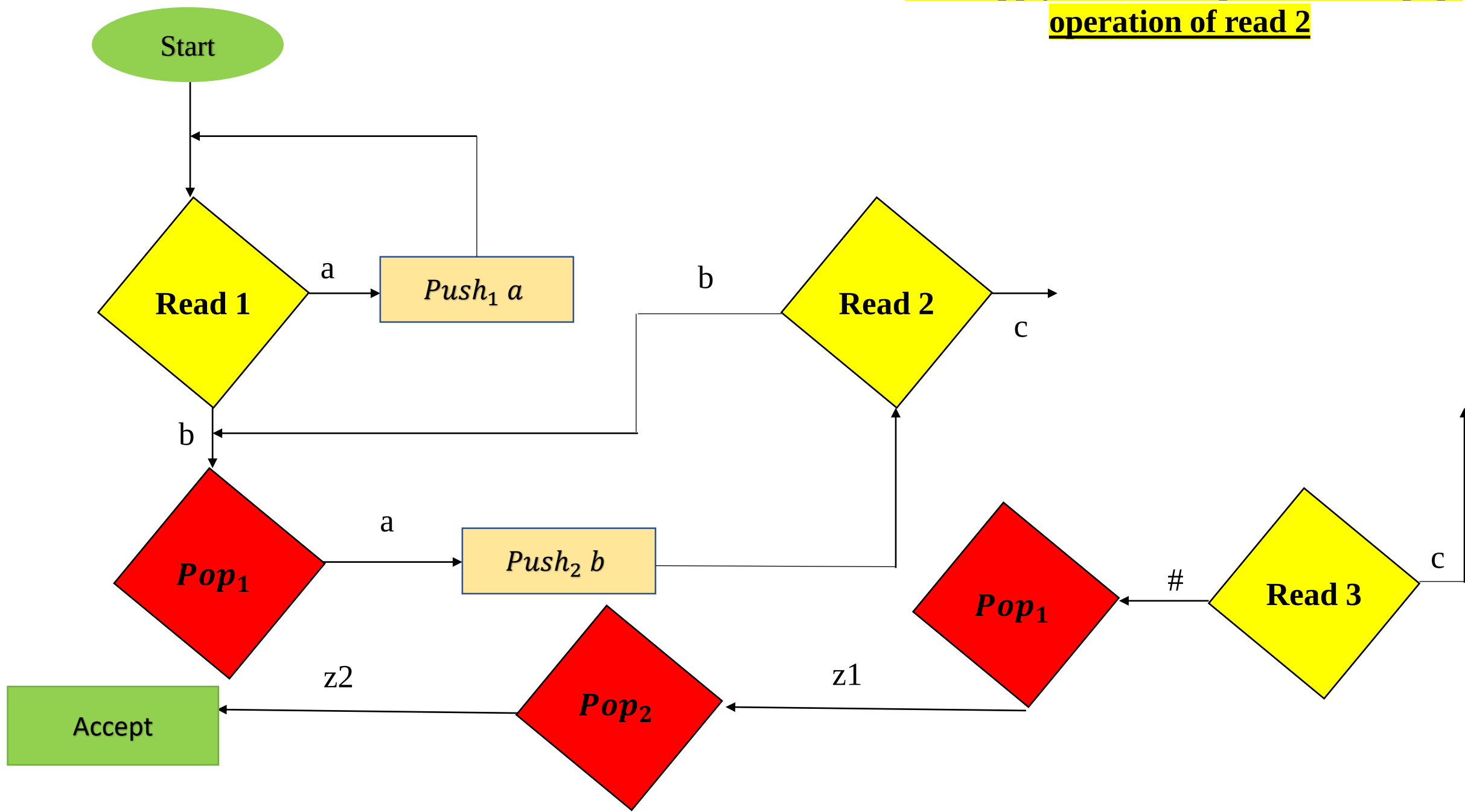
Draw all possibilities of READ



Now Apply the Push Operation and pop operation of read 1



Now Apply the Push Operation and pop operation of read 2



Now Apply the Push Operation and pop operation of read 2

